

THE OUTSIDE STORY

► **80% global photovoltaic production capacity concentrated in China**

► **Solar PV module imports fell 54% in FY26 over FY25**

► **Solar PV cell imports rise 86% in FY26 over FY25**

► **India module manufacturing capacity is over 172 GW**

► **Cell manufacturing capacity is 30 GW, implying huge import dependence**



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India imported photovoltaic cells valued at \$3,058.85 million in FY26, a jump of 86% over FY25, while PV module imports declined 53.77% to \$994.66 million during the same period, according to the commerce ministry data. The rise in PV cell demand was driven by a record 51 GW of solar deployment in FY26.

Cells are the building blocks of solar modules. The module manufacturing requires polysilicon, which is melted to form solid blocks of silicon known as ingots. These are sliced into wafers, serving as the foundational material

for solar cells.

This is where the Chinese dependence for the core lies. India has multiplied its solar installations tenfold in the past decade, yet it imports nearly all upstream essentials—98% of wafers and 100% of polysilicon. India has 2GW of ingot and wafer capacity and nothing of polysilicon.

Equipment bottlenecks are adding to the dependence. Over 90% of vital upstream tools, like Siemens reactors, ingot furnaces, diamond-wire saws, and other systems are imported, with Chinese firms controlling module as-

sembly lines, according to a recent paper on India's PV Manufacturing and Its Strategic Inflection Points by TERI.

Indian entities, such as HHV and Cliantech, on the other hand, offer only basic alternatives. This reliance amplifies foreign exchange risks, supply disruptions, and geopolitical tensions, especially in capital-heavy stages. Without targeted incentives and research & development, toolchains will remain the sector's silent saboteur, it says.

Vinay Thadani, Director & CEO, GREW Solar, says India's